String theory's proposal, depending on how hard you think about it, is either intuitively pleasing or thoroughly baffling. Since we speak of the "fabric" of spacetime, the suggestion goes, maybe spacetime is stitched out of strings much as a shirt is stitched out of thread. That is, much as joining numerous threads together in an appropriate pattern produces a shirt's fabric, maybe joining numerous strings together in an appropriate pattern produces what we commonly call spacetime's fabric. Matter, like you and me, would then amount to additional agglomerations of vibrating strings—like sonorous music played over a muted din, or an elaborate pattern embroidered on a plain piece of material—moving within the context stitched together by the strings of spacetime.

I find this an attractive and compelling proposal, but as yet no one has turned these words into a precise mathematical statement. As far as I can tell, the obstacles to doing so are far from trifling. For instance, if your shirt completely unraveled you'd be left with a pile of thread—an outcome that, depending on circumstances, you might find embarrassing or irritating, although probably not deeply mysterious. But it thoroughly taxes the mind (my mind, at least) to think about the analogous situation with strings—the threads of spacetime in this proposal. What would we make of a "pile" of strings that had unraveled from the spacetime fabric or, perhaps more to the point, had not yet even joined together to produce the spacetime fabric? The temptation might be to think of them much as we do the shirt's thread—as raw material that needs to be stitched together—but that glosses over an absolutely essential subtlety. We picture strings as vibrating in space and through time, but without the spacetime fabric that the strings are themselves imagined to yield through their orderly union, *there is no space or time*. In this proposal, the concepts of space and time fail to have meaning until innumerable strings weave together to produce them.

Thus, to make sense of this proposal, we would need a framework for describing strings that does not assume from the get-go that they are vibrating in a preexisting spacetime. We would need a fully spaceless and timeless formulation of string theory, in which spacetime emerges from the collective behavior of strings.

Although there has been progress toward this goal, no one has yet come up with such a spaceless and timeless formulation of string theory